

1

Interval: $(-1, 1)$

Parentheses mean endpoints are **not included**, so:

$$-1 < w < 1$$

Graph: open circle at -1 , open circle at 1 , shade between them.

2

(a)

$$64^{1/3} = \sqrt[3]{64} = 4$$

(b)

$$(-27)^{1/3} = \sqrt[3]{-27} = -3$$

(c)

$$\left(-\frac{1}{27}\right)^{1/3} = \sqrt[3]{-\frac{1}{27}} = -\frac{1}{3}$$

3

$$(6x - 9)(x^2 - x + 1)$$

Distribute:

$$\begin{aligned} & 6x(x^2 - x + 1) - 9(x^2 - x + 1) \\ &= 6x^3 - 6x^2 + 6x - 9x^2 + 9x - 9 \\ &= 6x^3 - 15x^2 + 15x - 9 \end{aligned}$$

4

$$\frac{x^2 - 81}{x^2 - 25} \cdot \frac{x + 5}{x + 9}$$

Factor:

$$\begin{aligned} x^2 - 81 &= (x - 9)(x + 9) \\ x^2 - 25 &= (x - 5)(x + 5) \end{aligned}$$

So:

$$\frac{(x-9)(x+9)}{(x-5)(x+5)} \cdot \frac{x+5}{x+9}$$

Cancel $x + 9$ and $x + 5$:

$$\frac{x-9}{x-5}$$

5

$$\frac{9}{x-9} - \frac{18}{x^2} = 0$$

Move one term:

$$\frac{9}{x-9} = \frac{18}{x^2}$$

Cross multiply:

$$9x^2 = 18(x-9)$$

Divide by 9:

$$x^2 = 2x - 18$$

$$x^2 - 2x + 18 = 0$$

Discriminant:

$$b^2 - 4ac = (-2)^2 - 4(1)(18) = 4 - 72 = -68$$

Since the discriminant is negative, there are no real solutions.

NO REAL SOLUTION

6

$$x(2x+5) \geq 0$$

Find critical points:

$$x = 0$$

$$2x + 5 = 0$$

$$x = -\frac{5}{2}$$

Test intervals:

For $x < -\frac{5}{2}$, both factors are negative, so product is positive.

For $-\frac{5}{2} < x < 0$, one factor is positive and one is negative, so product is negative.

For $x > 0$, both factors are positive, so product is positive.

Since the inequality is ≥ 0 , include the zeros:

$$\left(-\infty, -\frac{5}{2}\right] \cup [0, \infty)$$

7

Slope is $\frac{6}{7}$, y-intercept is 5.

Use slope-intercept form:

$$y = mx + b$$

$$y = \frac{6}{7}x + 5$$

8

$$f(z) = 5 - 4z^2$$

Given $z = -2$ and $z = 0$.

$$f(-2) = 5 - 4(-2)^2 = 5 - 4(4) = 5 - 16 = -11$$

$$f(0) = 5 - 4(0)^2 = 5$$

Net change:

$$f(0) - f(-2) = 5 - (-11) = 16$$

Average rate of change:

$$\frac{f(0) - f(-2)}{0 - (-2)} = \frac{16}{2} = 8$$

So:

$$\text{Net change} = 16$$

$$\text{Average rate of change} = 8$$

9

$$g(x) = x^2 - x - 20$$

Factor:

$$g(x) = (x - 5)(x + 4)$$

Zeros:

$$x = 5, \quad x = -4$$

The parabola opens upward because the coefficient of x^2 is positive.

Vertex:

$$x = \frac{-b}{2a} = \frac{-(-1)}{2(1)} = \frac{1}{2}$$

$$\begin{aligned} g\left(\frac{1}{2}\right) &= \left(\frac{1}{2}\right)^2 - \frac{1}{2} - 20 \\ &= \frac{1}{4} - \frac{1}{2} - 20 = -20.25 \end{aligned}$$

So the graph needs to show zeros at -4 and 5 , and the vertex near $(0.5, -20.25)$.

Best window:

$$[-7, 7] \text{ by } [-25, 20]$$

10

Match each parent function:

$$f(x) = x^2$$

This is the upward-opening parabola, so it matches **IV**.

$$f(x) = x^3$$

This is the S-shaped cubic graph, so it matches **II**.

$$f(x) = \sqrt{x}$$

This starts at $(0, 0)$ and curves rightward, so it matches **I**.

$$f(x) = |x|$$

This is the V-shaped graph, so it matches **III**.

Final answers:

$$x^2 : \text{IV}$$

$$x^3 : \text{II}$$

$$\sqrt{x} : \text{I}$$

$$|x| : \text{III}$$

11

$$P(x) = x^3 + x^2 - 20x$$

Factor out x :

$$P(x) = x(x^2 + x - 20)$$

Factor the quadratic:

$$x^2 + x - 20 = (x + 5)(x - 4)$$

So:

$$P(x) = x(x + 5)(x - 4)$$

Zeros:

$$x = 0, \quad x = -5, \quad x = 4$$

Answer with zeros:

$$-5, 0, 4$$

Since the leading term is positive x^3 , the graph goes down on the left and up on the right. It crosses the x-axis at -5 , 0 , and 4 . Correct graph: the first graph shown.

12

Continuous compound interest formula:

$$A = Pe^{rt}$$

Here:

$$P = 1000, \quad r = 0.065$$

(a) For $t = 3$:

$$A = 1000e^{0.065(3)}$$

$$A = 1000e^{0.195}$$

$$A \approx 1215.31$$

$$\boxed{\$1215.31}$$

(b) For $t = 6$:

$$A = 1000e^{0.065(6)}$$

$$A = 1000e^{0.39}$$

$$A \approx 1476.98$$

$$\boxed{\$1476.98}$$

(c) For $t = 18$:

$$A = 1000e^{0.065(18)}$$

$$A = 1000e^{1.17}$$

$$A \approx 3221.99$$

$$\boxed{\$3221.99}$$

13

Terminal point:

$$P(x, y) = \left(-\frac{3}{5}, -\frac{4}{5}\right)$$

On the unit circle:

$$\cos(t) = x$$

$$\sin(t) = y$$

So:

$$\boxed{\sin(t) = -\frac{4}{5}}$$

$$\boxed{\cos(t) = -\frac{3}{5}}$$

$$\tan(t) = \frac{\sin(t)}{\cos(t)}$$

$$\tan(t) = \frac{-\frac{4}{5}}{-\frac{3}{5}}$$

$$\tan(t) = \frac{4}{3}$$

$$\boxed{\tan(t) = \frac{4}{3}}$$

14

From the triangle, use the $45^\circ - 45^\circ - 90^\circ$ triangle relationship.

In a $45^\circ - 45^\circ - 90^\circ$ triangle:

$$\text{hypotenuse} = (\text{leg})\sqrt{2}$$

The given leg is 34, so:

$$x = 34\sqrt{2}$$

$$\boxed{x = 34\sqrt{2}}$$

15

Expression:

$$\frac{\cos(u) \sec(u)}{\tan(u)}$$

Use the reciprocal identity:

$$\sec(u) = \frac{1}{\cos(u)}$$

Substitute:

$$\frac{\cos(u) \cdot \frac{1}{\cos(u)}}{\tan(u)}$$

Cancel:

$$\frac{1}{\tan(u)}$$

Use reciprocal identity:

$$\frac{1}{\tan(u)} = \cot(u)$$

$$\boxed{\cot(u)}$$

16

Given:

$$u = 4i + j$$

$$v = i - 3j$$

$$w = 3i + 2j$$

Write as vectors:

$$u = \langle 4, 1 \rangle$$

$$v = \langle 1, -3 \rangle$$

$$w = \langle 3, 2 \rangle$$

Find $u \cdot v$:

$$\begin{aligned} u \cdot v &= (4)(1) + (1)(-3) \\ &= 4 - 3 = 1 \end{aligned}$$

Find $u \cdot w$:

$$\begin{aligned} u \cdot w &= (4)(3) + (1)(2) \\ &= 12 + 2 = 14 \end{aligned}$$

Now multiply:

$$(u \cdot v)(u \cdot w) = (1)(14) = 14$$

$$\boxed{14}$$

17

$$\sum_{k=6}^9 k(k+3)$$

Write out each term from $k = 6$ to $k = 9$:

$$\begin{aligned} &6(6+3) + 7(7+3) + 8(8+3) + 9(9+3) \\ &= 6(9) + 7(10) + 8(11) + 9(12) \\ &= 54 + 70 + 88 + 108 \end{aligned}$$

$$= 320$$

So without sigma notation:

$$6(9) + 7(10) + 8(11) + 9(12)$$

Simplified value:

$$320$$

   ...  Sources